

Agile vs. Waterfall: A Comparative Study in IT Project Success Rates

Kaustubh Deotighare
Computer Technology
Yeshwantrao Chavan College Of Engineering
Nagpur, India
deotigharekaustubh@gmail.com

Soham Bakane
Computer Technology
Yeshwantrao Chavan College Of Engineering
Nagpur, India
sohambakane3@gmail.com

Abstract—Project management methodologies significantly influence IT project success. Agile and Waterfall project management are two of the most widely used project management methodologies in the industry, each with benefits and drawbacks. This research paper presents a critical comparison of Agile and Waterfall methodologies as they relate to the success of IT projects. Through a literature review and analysis of recent surveys of IT professionals, this study explores several measures of project success, including whether the project is complete, whether the project was completed on time and within budget, the extent to which the project delivered 'as advertised', and customer satisfaction after a project's completion using Agile or Waterfall methodologies. Generally, Agile methodologies lead to higher success rates, especially when there is rapid change, high complexity, and an emphasis on client satisfaction. Waterfall may serve better as a project methodology with clearly defined, stable requirements, and lower levels of uncertainty. This paper seeks to simply present the findings to help IT project managers decide on the correct methodology to help ensure a satisfactory project outcome and improve success rates.

Keywords—Agile, Waterfall, IT Project Management, Project Success Rates, Software Development Methodologies, Comparative StudyIntroduction

I. INTRODUCTION

As the information technology (IT) industry continues to evolve, effective project management is vital for the efficient delivery of software products and services. Project success is largely based on the criteria for success, which can vary depending on stakeholders and goals. However, success criteria quite often include variables such as time, budget, scope, and customer satisfaction. The methodology or framework of project management selected will influence all of these variables, especially success.

Agile and Waterfall are two of the most common approaches to consider when selecting a project management methodology. The Waterfall approach does follow a very linear flow of project, meaning that project phases cannot be completed until previous phases are completely finished. Waterfall is appropriate for projects with relatively well-defined requirements and little anticipated change. Agile is helpful for projects with a complex and emerging set of requirements, and therefore it leverages ongoing collaboration with the customer, facilitating many iterations of the project to develop an evolving series of project deliverables. This has been very helpful for the IT industry considering only 16% of software projects were considered successful in 2010. After implementations, if users only use 16% of the software or service features provided by the IT

team, the 0% success rate would stand to reason, and this is where Agile can afford the customer success.

While both methodologies have been effective in various contexts, it is still a topic of study and interest to understand which methodology can produce a higher chance of a successful project. This paper seeks to accomplish a comparative analysis of Agile and Waterfall methodologies through an exploration of their strengths, weaknesses, and influence on project success rates in the IT industry. The aim of the study is to generate informative findings for IT project managers and decision-makers, whom can make meaningful choices in the methodology selection process and ultimately improve the likelihood of a successful delivery of the project.

II. RELATED WORK

Project management methods are important when it comes to bringing about project outputs in the IT domain. There is a growing body of literature that compares the success of projects that use Agile versus Waterfall project management methodologies.

A. Agile Methodology

Agile methodology adopts an iterative and incremental approach to development, focusing on flexibility, frequent inspection and adaptation, and close collaboration between developers and stakeholders. Agile encourages adaptive planning, evolutionary development, early delivery, and continuous improvement using methods such as Scrum, Kanban, and Extreme Programming (XP) [1]. There are also studies comparing Agile projects and success whereby projects using Agile consistently report greater customer satisfaction because Agile projects typically deliver workable increments at regular intervals, while being able to adapt quickly to changes [2].

B. Waterfall

In project management, the Waterfall methodology is a traditional, linear sequence of project activities, such as requirements, design, implementation, testing, deployment, and maintenance, where each activities is only completed before going to the net [4]. The linear sequence of activity is very well suited for projects where requirements need to be clear and not flex or change.

Waterfall is used extensively in projects that require strict regulatory compliance and where predictable outcomes are paramount to the success of the project. Lots of research has suggested that Waterfall projects can be inefficient if the projects requirements change or are largely unclear [5]. Based on the Standish Group Chaos Report [6], projects that are strictly Waterfall can often expect late delivery and scope

"creep", particularly if the project environment is highly fluid like IT and other nascent fields.

C. Measuring Project Success

Project success is typically evaluated by various key components such as the project's delivery on time, followed its budget, fulfilled its scope, delivered the deliverables to a quality standard and satisfied the customer [7]. There are numerous studies that show that Agile produced better customer satisfaction and more flexibility in project progress over Waterfall, while Waterfall provided a higher degree of predictability and structured plan [8].

D. Measuring Project Success

Research topics comparing the methodologies of Agile and Waterfall are frequent in research literature; however, much of the research is outdated or heavily driven by specific industries. There is a gap for updated research into the comparison of the methodologies of Agile and Waterfall that considers the complexity of today's IT projects and reports contemporary success rates. This research aims to fill that gap, by undertaking a comparison of the two methodologies, specifically from both literature and previously reviewed projects.

III. METHODOLOGY

In this study, we employed a combination of literature review and survey data collection techniques, a mixed-method research design, to compare the reported success rates of IT projects that utilized either the Agile or Waterfall methodologies.

E. Research Design

The study consisted of two phases:

Phase 1: A systematic literature review of existing studies, industry reports, and white papers reporting on Agile and Waterfall project success rates.

Phase 2: Primary source data collection via an online survey that targeted IT professionals who had experience with Agile and Waterfall projects.

F. Data Collection

We designed and issued a structured survey to IT project managers, developers, quality assurance professionals, and business analysts across a variety of organizations. The survey was intended to capture information regarding:

- Methodology (Agile or Waterfall) used for the project
- Project success rates – on-time delivery, met budget, satisfactory to client
- Type of industry and project size
- Common barriers or challenges issues experienced during the course of the project
- We received and analyzed 150 valid responses.

C. Data Collection

The survey data is analyzed using descriptive statistics and calculated the percentage of success rates for Agile and Waterfall projects. Comparative analysis was performed using graphical visualization to show the differences in key indicators of project success:

- On-time delivery

- Budget
- Customer / end-user satisfaction

D. Success Criteria

For this study, we define project success when all the following criteria are met:

- Delivered on time
- Delivered within budget
- Had the project scope met as defined
- Satisfied the end-user / customer.

The data that were gathered from survey responses were analyzed to compare the project success rates for Agile versus Waterfall using key project performance measures. The key measures included were on-time delivery, budget and customer satisfaction.

TABLE I.

Features	Agile	Waterfall
Process Flow	Iterative & Incremental	Linear & Sequential
Flexibility	Highly Flexible	Rigid, Low Flexibility
Customer Involvement	Continuous	Limited (at milestones)
Change Management	Easily Accommodated	Difficult to Implement
Project Success Rate	Generally Higher	Generally Lower
Best Fit For	Complex, Dynamic Projects	Simple, Well-Defined Projects

IV. RESULT

The data retrieved from the survey responses were reviewed as to the success rates of Agile and Waterfall compared across key project performance metrics. The key metrics evaluated were on-time delivery, on-budget delivery, and customer satisfaction. Authors and Affiliations

E. Success Rate Comparison

- Agile Projects: 80 projects
- Waterfall Projects: 70 projects

B. Visual Representation: Success Rate Bar Chart

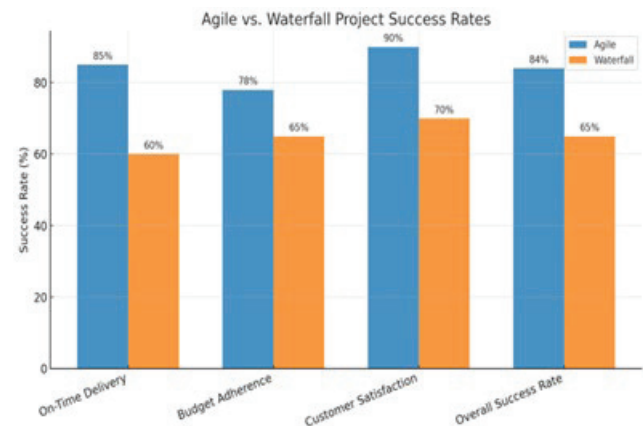


Fig. 1. Visual Representation: Success Rate Bar Chart

C. Examination of Result

The data shows that Agile projects consistently performed better than Waterfall projects regarding all scores:

- **Timeliness:** More Agile projects were delivered on time (85%) than Waterfall projects (60%).
- **Budget:** Agile projects were more likely to come in on budget.
- **Client Satisfaction:** Because Agile projects had continuous customer involvement, the customer satisfaction rate was greater than the Waterfall project (90% vs. 70%).
- **Overall Success Rate:** Agile projects overall, achieved a higher success rate(84%) than Waterfall projects realized(65%).

The above data suggests that Agile project scenarios retained strong value for use in a rapidly changing and fluid project context.

V. DISCUSSION

The comparison provided via the perspectives of this research study contributes insightful knowledge about the success of IT projects attained through either the Agile or Waterfall methods. Therefore, based on the findings of the survey and literature review, it is evident that Agile methods consistently outperform Waterfall methods based on several definitions of success, particularly on time delivery, on budget delivery, and customer satisfaction.

D. Interpretation of Results

The success of Agile projects can likely be linked to many contributory factors, which include:

Customer Collaboration : Agile promotes communication and collaboration with customers, which allows for better definition and understanding of changing requirements and minimizes the chance of failure due to mismatched expectations.

Iterating Development: Agile allows for a speedier timeline for the delivery of functional products, which exposes any issues earlier in the project timeline allowing for quicker fixes.

Flexibility: Agile has a considerable amount of flexibility which means that teams can manage scope, new technology, and shifting customer demands all while allowing the project team to maintain quality on-time delivery and reported customer satisfaction.

Conversely, Waterfall projects are generally considered superior when the project requirements are clear and change resistant. Waterfall's documentation and process management has an advantage when projects are conducted in heavily regulated environments.

E. Situational Fit

The research further emphasizes that no methodology is constantly better. Agile and Waterfall should be determined based on:

Project Complexity: Agile works best for more complex and dynamic projects, Waterfall fits simpler project types need more stability and predictability should fit simpler project types.

Requirements Stability: Waterfall would be the proper method when the project has stable, predictable requirements, and less engagement with the client.

Team and Organizational key performance indicators: Organizations that have the key performance indicators, frameworks and rituals to observe when implementing Agile can measure their value above Waterfall. Newer teams to Agile may to experience more barriers in this situation than Agile's benefits

F. Study Limitation

Although the findings from the study can add value, this study has limitations:

- The survey sample size was limited to 150 respondents, although the respondents represented a range of diversity in size, skill level, knowledge and anecdotal recording.
- The study focused on IT projects. The findings may not be generalizable to projects in non-IT disciplines.
- The success rates were based on self-reported success rates, and could be influenced by respondent bias.
- Future research has many opportunities to improve on this project such as an expanded sample size, and including comparative cross-industry data and project data from real-world case studies.

VI. CONCLUSION

This research paper has completed a comparative study of the Agile and Waterfall project management methodologies on the success rates of IT project management. From the literature review and the prompt survey-based primary data collected, it is clear that Agile methodology provides greater overall success rates than Waterfall methodology, particularly for projects that are on time, within budget, and meet customer satisfaction.

The iterative development process of Agile methodology including its flexibility and relevance to project stakeholders business about development and the evolving requirements for projects that lead to success is well documented. On the other hand, the Waterfall model can be very attractive for projects that have well-defined and stable requirements and require minimal changes to get to a successful project completion.

This study suggests that acceptance of a project methodology is not an appropriate decision purely by the industry trend but, more so through the evaluation of the complexity of a project, the stability of the requirements, team capabilities, and corporate culture. Both methodologies have strengths and weaknesses, but if aligned in accordance with the project confines than they both have prospects for project success.

Future research could examine hybrid models based on Agile-waterfall, hybrid models, or provide more extensive case studies across a broad range of industries to improve upon these findings.

This study contributes to the growing body of knowledge that can assist IT project managers in making informed, data-driven decisions when selecting a project management methodology.

REFERENCES

- [1] L. T. Khoza and C. Marnewick, "Waterfall and Agile information system project success rates – a South African perspective," *South African Computer Journal*, vol. 32, no. 1, pp. 43–73, Jul. 2020.
- [2] G. Sirisha, V. Sarada, J. E. P., B. Haralayya, B. V. RamaKrishna, and D. Bisen, "Project Management Methodologies: A Comparative Analysis of Agile and Waterfall Approaches," *Library Progress (International)*, vol. 44, no. 3, pp. 17237–17246, Jul.–Dec. 2024.
- [3] T. J. Lehman and A. Sharma, "Software Development as a Service: Agile Experiences," presented at SRII Conference, IBM Research, Aug. 2011.
- [4] R. Shaydulin and J. Sybrandt, "To Agile, or not to Agile: A Comparison of Software Development Methodologies," *arXiv preprint*, Apr. 2017.
- [5] A. Perez Veiga, "Project Success in Agile Development Projects," *arXiv preprint*, Nov. 2017.
- [6] A. B. M. Moniruzzaman and S. Akhter Hossain, "Comparative Study on Agile Software Development Methodologies," *Global Journal of Computer Science and Technology*, vol. 13, no. C7, pp. 5–18, Mar. 2013.
- [7] P. Abrahamsson, N. Oza, and M. T. Siponen, "Agile Software Development Method, A Comparative Review," *arXiv preprint*, Mar. 2019.
- [8] C. S. Maharao, "A Comparative Analysis of Agile, Waterfall, and Hybrid Methodologies in Software Project Success," *ShodhKosh*, vol. 3, no. 2, pp. 917–926, Dec. 2022.
- [9] S. R. Kathram and S. R. K. Nersu, "Waterfall vs. Agile: A Comparative Study on Project Success in Hybrid Methodologies," *Int. J. Mach. Learn. Res. Cyber Secur. Artif. Intell.*, vol. 12, no. 1, pp. 510–530, Dec. 2021.
- [10] A. J. Gilbert Silvius, "Waterfall and Agile information system project success rates – a South African perspective," *J. Cleaner Prod.*, Aug. 2017.